RIGOLO BAUGE ROBIN | RESUME

Vodafone: (675) 81233406
rigobauge@gmail.com

Software Engineer - Skills/Expertise Portfolio

June 17, 2026

" You Need NO Fancy Qualification nor an Expensive Education to Accomplish Wonders ! "

- Full Stack Web Developer
- Database (Data Analysis & Dashboard Developer)
- Educator- Lecturer / Trainer / Teacher
- Advanced Vector Digital Arts Graphic Designer

Current Engagement: Systems Achitech & Software Engineer engaged by Coffee Industry Corporation; now National Coffee Authority @ Lae

CV REMEDIAL PAGES

REMEDIAL INFORMATION - SOFTWARE PROJECTS DESCRIPTION SUMMARY

These pages contain the Information of the projects that I, Rigolo Bauge Robin, Systems Architect and Software Engineer have developed. The purpose of these remedial pages is to give more insights of the developer's extensive and comprehensive knowledge and hands-on software development expertise.

Executive-Level Project Summary

Project: PNG Curriculum OS

Comprehensive Project Intelligence Analysis and Professional Portfolio Summary

1. Executive Summary

PNG Curriculum OS is a massive, enterprise-grade, AI-powered Learning Management System (LMS) developed for the educational ecosystem of Papua New Guinea. Far exceeding the scope of a traditional LMS, it serves as an autonomous educational engine that manages the entire lifecycle of curriculum generation, interactive student tutoring, risk assessment, and school administration. Comprising over 24 interconnected Django applications, 240+ Python modules, and an intricate 37-service AI layer, the platform bridges the gap between static curriculum standards and dynamic, personalized learning paths.

2. Project Description

Built to revolutionize education accessibility and quality, PNG Curriculum OS transforms raw curriculum standards (often ingested from PDFs) into fully interactive, personalized learning experiences. The system supports multi-role access (Students, Teachers, Administrators, Parents) and orchestrates everything from automated lesson plan generation and adaptive AI tutoring to proactive dropout risk detection. By combining large language models (via OpenRouter), vector databases (pgvector), and a robust async task processing architecture (Celery/Redis), the platform acts as an always-on "AI Teacher" and "Admin Assistant," capable of scaling national educational frameworks to individual student needs.

3. Technical Architecture Overview

The system employs a deeply modular, service-oriented architecture centered around Django, enhanced by asynchronous task queues and complex AI pipelines.

* ***Core Framework***: Django MVC monolith structured into 24 highly cohesive apps (e.g., `academics`, `student_management`, `ai_engine`, `vault`).
* ***Asynchronous Processing***: Celery and Redis manage background tasks, heavily utilized for cron-based risk assessments (`RiskEngine`), attendance anomaly detection, and intensive AI curriculum generation tasks.
* ***AI Integration Layer***: A dedicated `ai_engine` with 37 micro-services handles LLM orchestration. It utilizes OpenRouter for LLM inference (e.g., GPT-4o-mini), while local `sentence-transformers` and `pgvector` manage semantic search and embedding storage.
* ***Frontend Strategy***: A hybrid approach utilizing Django templates enhanced with HTMX for reactive backend-driven UI components, complemented by React/Redux (`TeacherAILessonGenerator.tsx`, `authSlice.ts`) for complex, highly interactive client-side views.
* Infrastructure: Containerized with Docker and Docker Compose, served via Gunicorn and NGINX, prepared for scalable cloud deployment.

4. Technology Stack

* ***Backend Engineering***: Python 3.12+, Django 6.0.3, Django REST Framework, Celery, Redis.
* ***Artificial Intelligence & Machine Learning***: OpenRouter API, LangChain, LangGraph, HuggingFace (`sentence-transformers`), PyMuPDF (PDF parsing).
* ***Databases & Storage***: PostgreSQL, `pgvector` (Vector Database), AWS S3 (`boto3`, `django-storages`).
* ***Frontend***: React, TypeScript, Redux Toolkit, TailwindCSS, HTMX.

* ***DevOps & Deployment***: Docker, NGINX, Gunicorn, Shell scripting.

5. Key Features and Functionalities
* ***AI Autonomous Teaching Agent***: Provides step-by-step interactive lessons with real-time feedback, adapting to student performance dynamically.
* ***End-to-End Curriculum Generation***: Ingests raw MOE PDFs, parses them via AI, and automatically generates structured syllabi, lesson plans, and assessments.
* ***Semantic Knowledge Graph***: Maps prerequisites and topic clusters utilizing vector embeddings, ensuring students learn concepts in logical progression.
* ***Predictive Risk & Intervention Engine***: Background Celery workers constantly evaluate student attendance and grades, triggering automated early warnings and custom intervention plans for at-risk students.
* ***Multi-Role Dashboards***: Distinct, data-rich interfaces for Students (interactive learning), Teachers (auto-grading, insights), Admins (school-wide risk analytics), and Parents (academic tracking).

6. Engineering Challenges Solved
* ***Complex LLM Orchestration***: Designing a robust pipeline that breaks down massive curriculum generation tasks into sequential, verifiable steps to avoid LLM hallucination and timeout limits.
* ***Vector Search & Data Synchronization:*** Implementing a hybrid search strategy using `pgvector` to match student queries with exact curriculum chunks (Cosine Distance), bridging the gap between natural language questions and rigid syllabus structures.
* ***Proactive System Architecture***: Shifting from a reactive web app to a proactive one by implementing signal-driven and Celery-scheduled engines (`RiskEngine`, `AttendanceAI`) that identify student failure risks before they happen.
* ***State Management in Interactive AI***: Maintaining context across long-running AI tutoring sessions using sophisticated database modeling (`StudentTopicProgress`, `TopicRelationship`).

7. Innovation and Technical Creativity
* ***Deterministic Knowledge Graph via AI:*** Successfully combining the creative output of LLMs with strict, deterministic database constraints to build a reliable educational Knowledge Graph without logical loops or missing prerequisites.
* ***Automated National Exam Simulation***: A pipeline that cross-references the entire curriculum database to programmatically construct realistic, balanced national exam papers.
* ***Algorithmic Intervention Generation***: Rather than just flagging a student as "at risk," the system automatically constructs an `InterventionPlan` with actionable tasks for teachers and parents.

8. Developer Competency Showcase
Based entirely on the codebase implementation, the developer demonstrates exceptional capability in:

* ***System Architecture***: Ability to design and structure a massive 24-app Django project without losing cohesion. Excellent separation of concerns (e.g., separating AI logic into isolated `engines` rather than polluting Django views).
* ***Advanced AI Engineering***: Deep understanding of practical AI application—moving beyond simple chat wrappers to implement RAG (Retrieval-Augmented Generation), vector embeddings, and autonomous agent workflows.
* ***Asynchronous & Background Processing***: Proficiency with Celery/Redis for managing heavy background workloads, a critical skill for scaling production applications.
* ***Full-Stack Proficiency***: Comfort working across the stack, from complex PostgreSQL extensions (`pgvector`) to Python backend logic, to modern TypeScript/React frontend components.
* ***Product Thinking***: Designing features not just for technical elegance, but for real-world impact (e.g., the `AttendanceAI` and `RiskEngine` explicitly designed to prevent student dropout).

9. Business Value and Organizational Impact
This system directly addresses major educational challenges in developing regions by democratizing access to high-quality, personalized tutoring. By automating curriculum generation and grading, it dramatically reduces teacher workload, allowing educators to focus on high-impact interventions. The predictive risk engines provide school administrators with the data needed to prevent student dropouts, ultimately improving national educational outcomes.

10. Recruiter-Focused Portfolio Summary
Project: PNG Curriculum OS — Enterprise AI Educational Platform
Role: Lead Full-Stack / AI Engineer
Tech: Python, Django, Celery, PostgreSQL (pgvector), React, TypeScript, OpenRouter API.

Developed a comprehensive, AI-driven Learning Management System designed to scale national curricula. Architected a 24-module Django monolith integrating 37 custom AI microservices. Engineered an automated curriculum ingestion pipeline that converts PDFs into interactive, vector-mapped learning paths using LLMs and pgvector. Implemented proactive background risk-assessment engines (Celery/Redis) that analyze attendance and grades to auto-generate teacher intervention plans. Demonstrated deep expertise in RAG, system architecture, async task processing, and full-stack development, delivering a product that transitions traditional education into adaptive, personalized learning.

11. Resume Version (150–250 Words)
Lead Engineer | PNG Curriculum OS
* Architected and developed an enterprise-grade, AI-powered Learning Management System consisting of 24 interconnected Django applications and 37 custom AI services.*
* Engineered an automated curriculum pipeline that ingests raw educational standards (PDFs) and utilizes LLMs (OpenRouter) to generate structured lesson plans, assessments, and national exam simulations.
* Implemented a RAG-based AI Tutor and Semantic Search using PostgreSQL (`pgvector`) and local sentence-transformers, enabling context-aware, 24/7 interactive student tutoring.
* Designed and deployed proactive background processing engines via Celery and Redis (`RiskEngine`, `AttendanceAI`) that analyze real-time student data to detect dropout risks and automatically generate actionable intervention plans.
* Built a hybrid frontend using Django templates, HTMX, and React/TypeScript for highly interactive, role-based dashboards (Student, Teacher, Admin, Parent).
* Established a robust, scalable architecture utilizing Docker, Gunicorn, and NGINX, demonstrating full-stack proficiency from database schema design to complex LLM orchestration.

12. LinkedIn Version
🚀 Just wrapped up a massive milestone on PNG Curriculum OS—a comprehensive, AI-powered educational platform designed to transform learning!

Instead of building just another LMS, I architected an autonomous teaching engine. The system ingests raw curriculum PDFs and uses LLMs to generate complete, interactive learning paths.

Highlights of the build:
🧠 ***37 Custom AI Services***: Orchestrating complex LLM pipelines for curriculum generation and real-time student tutoring.
🔍 ***Vector Search & RAG***: Implemented `pgvector` and semantic embeddings to map prerequisites and power an intelligent Knowledge Graph.
⚙️ ***Proactive Analytics:*** Built Celery-driven asynchronous engines (`RiskEngine`, `AttendanceAI`) that constantly analyze grades and attendance to predict dropout risks and auto-generate intervention plans.
💻 ***Full-Stack Stack***: Django, Python, PostgreSQL, Redis, Celery, React, TypeScript, and TailwindCSS.

It was an incredible engineering challenge to bridge the gap between static curriculum standards and dynamic, personalized AI education. Excited for the impact this will have!

#SoftwareEngineering #AI #Django #Python #EdTech #FullStack

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